

The Fabric of the Internet and AI

Lesson 1: Introduction to the Internet

Resources

The Fabric of the Internet and AI - Unit Guide

Lesson 1: Introduction to The Internet

Vocabulary

Word	Definition	Example/Image/Connection
Routers		
Switch	Helps send data to the right place inside a local network	
Internet Tour		
Server	Store and deliver websites, files, and messages when requested	

Networks - Free Sketch (Optional)

Use this space to sketch or jot down key ideas about how networks function. Think about routers, switches, servers, or any other parts of a network that stood out to you.

Question of the Day

How does the Internet connect billions of devices across the globe?

Lesson 2: Sending Bits

Vocabulary

Word	Definition	Example/Image/Connection
Transmission Control Protocol (TCP)	a protocol that ensures that all parts of a message are received and ordered correctly	

Binary Transmission - Free Sketch (Optional)

Use this space to sketch or jot down ideas about how binary data travels across a network. How do bits get from one device to another?

Question of the Day

How are messages broken into bits and sent across networks?

Lesson 3: IP Addresses

Vocabulary

Word	Definition	Example/Image/Connection
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IP Address		192.168.20.1  192.168.20.31 
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IP Addresses in Action - Free Sketch (Optional)

Feel free to use the space below to doodle, jot down thoughts, or sketch out how IP addresses are used in everyday life. You might map out how devices in your home, school, or a public space connect to a network using unique IP addresses.

Question of the Day

Why do devices on the Internet need addresses?

Lesson 4: Routers and Redundancy

Redundancy in Action - Free Sketch (Optional)

Use the space below to brainstorm or sketch examples of redundancy in everyday life. Think about how backup plans work in technology, transportation, or even nature. How does having multiple routes, backups, or fail-safes improve reliability?

Question of the Day

How does data find its way across the Internet and how does redundancy improve reliability?

Lesson 5: Packets

Vocabulary

Word	Definition	Example/Image/Connection
User Datagram Protocol (UDP)	A protocol that focuses on speed and does not guarantee all packets arrive. • Sends all of the packets without verifying their order or if they are all sent. • Useful when split seconds matter.	

Packet Transmission - Free Sketch (Optional)

Use this space to create a flowchart or diagram showing how data is broken into packets, sent, and reassembled. What happens if packets arrive out of order?

Question of the Day

How does data travel reliably to reach its destination on an unreliable network?

Lesson 6: DNS and HTTP

Vocabulary

Word	Definition	Example/Image/Connection
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Domain Name System (DNS)	Translates human-friendly website names (like www.example.com) into numerical IP addresses (and vice versa).	
Hypertext Transfer Protocol (HTTP)		

DNS Hierarchy - Free Sketch (Optional)

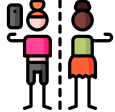
Use this space to create a flowchart showing how DNS helps your device find websites. How does your computer know where to go when you type in a web address?

Question of the Day

How do websites find their way to your browser using DNS and HTTP?

Lesson 7: The Internet's Hidden Barriers

Vocabulary

Word	Definition	Example/Image
Net Neutrality	The idea that all websites & online services should be treated equally by Internet providers. This means no site is slowed down or blocked, & companies can't pay for faster access	
Digital Divide		

Digital Divide Observations

Data	Who has the most access?	Who has the least access?	What surprised you?
Internet Use Over Time			
Home Broadband Use By Income			
Internet Use by Community Type			
Internet Use by Age			
Smartphone Dependency by Education			

Question of the Day

Who has access to the Internet, and who is left behind?

Lesson 9: Network Security

Vulnerabilities

Note: Don't forget to click the  icon to learn about the specific vulnerability you are investigating.

Phishing Email

Vulnerability Details	Observations, Thoughts, Questions	Mitigation Ideas
What kind of attack or weakness is this? Describe it briefly. Fake email pretending to be from IT asking for login info.	What happened in the scenario? What stood out to you? The email used urgent language to trick the employee. It looked real but had a suspicious link.	What would stop or reduce the impact of this vulnerability? Train employees to spot phishing, flag suspicious emails, and use email filters.



Weak Passwords

Vulnerability Details	Observations, Thoughts, Questions	Mitigation Ideas



Default Credentials

Vulnerability Details	Observations, Thoughts, Questions	Mitigation Ideas
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Social Engineering

Vulnerability Details	Observations, Thoughts, Questions	Mitigation Ideas

Question of the Day

What vulnerabilities exist in networks, and what strategies can mitigate their impact?

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Lesson 10: Emerging Technologies

AI vs Traditional Security - Free Sketch (Optional)

Use this space to visually compare AI-based and traditional security methods. How does AI improve security, and what challenges might it have?

Question of the Day

What makes AI effective at enhancing network security, and how does it compare to traditional methods?

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Lesson 3: IP Addresses

Resources

Activity Guide - IP Addressing and Visualizing

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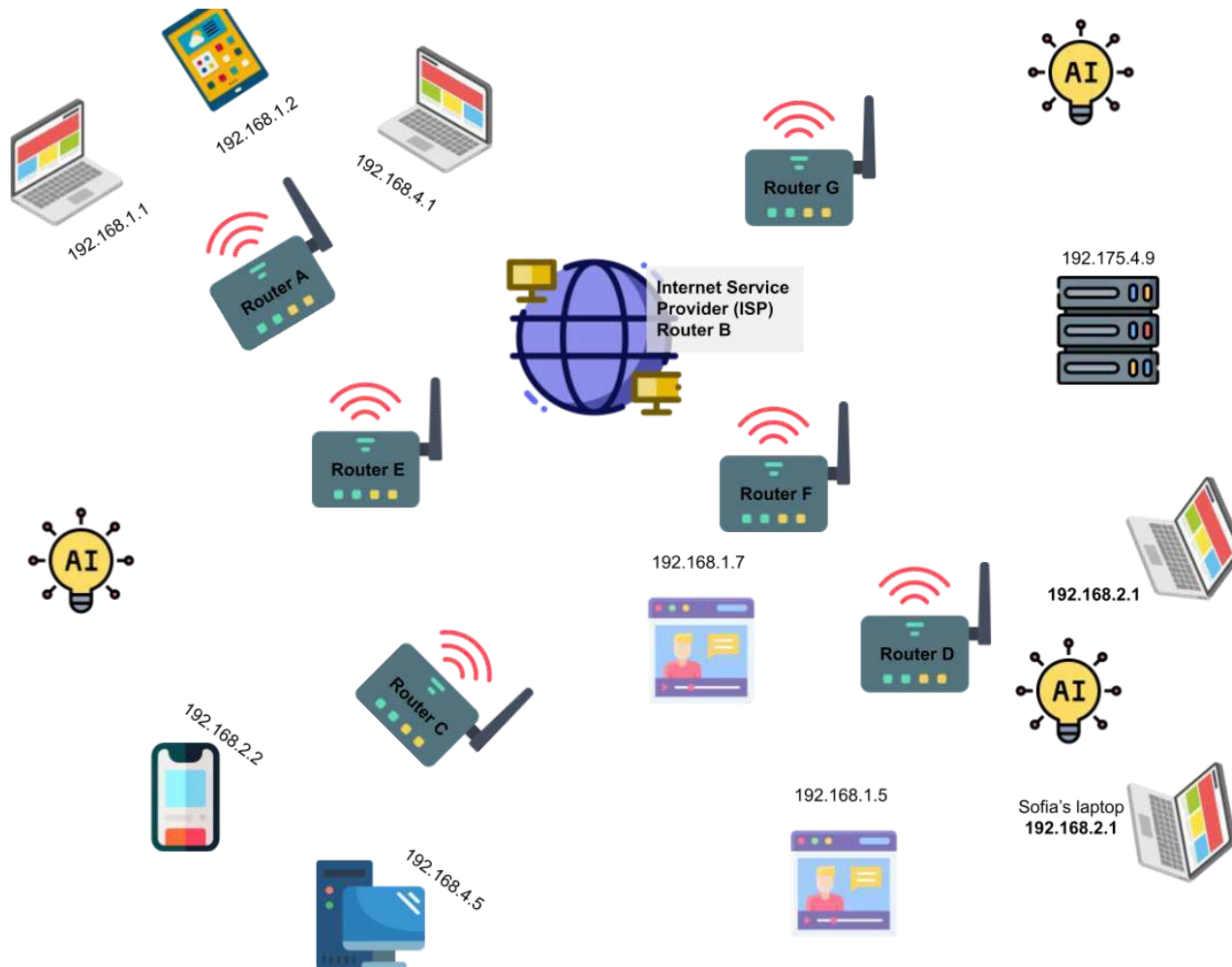
Network Map

Review the routing table below and the network diagram. Use the Device Type and Destination IP Address to label each device in the network diagram.

Device Type	Destination IP Address	Next Hop Router
Jordan’s Laptop	192.168.1.1	Router A
Mia’s Tablet	192.168.1.2	Router A
Ava’s Smartphone	192.168.2.2	Router C
Liam’s Laptop	192.168.4.1	Router A
Emily’s Streaming Device	192.168.1.5	Router D
Streaming Video Provider's Server	192.175.4.9	Router G
Ryan’s Laptop	192.168.2.1	Router D
Jake’s Streaming Device	192.168.1.7	Router F

Address Problem Solving

Look at the scenarios on Levels 1 through 4. After reading each scenario, use the routing table and network diagram to determine the possible path(s) for data transmission.



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Lesson 4: Routers and Redundancy

Resources

Network Diagram Analysis – Activity Guide

Part 1: Getting Started

1. Connect & Record Your Information
 - Log into the Internet Simulator.
 - Connect to a router that is different from your elbow partner's.
 - Record your assigned IP Address and Router Number in the space below:
 - My IP Address: _____
 - My Router Number: _____
2. Send Messages
 - You can write about:
 - What is your favorite food?
 - What is your favorite type of animal?
 - What is your favorite color?

Part 2: Diagramming Your Network

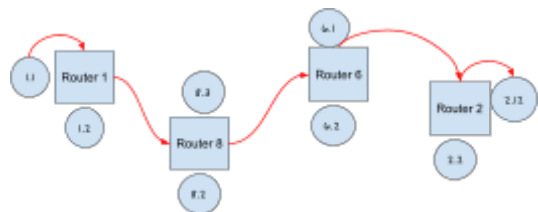
Instructions:

- Make a diagram of the path your messages take. Diagram up to three messages.
- In the Router tab, select “Show traffic to me” and “Show all routers”.
- Use the router log to see how the messages travelled between routers.
- Label your diagram clearly with:
 - Routers: Represent each router as a square. Label each with its router number.
 - Nodes: Represent each device with a circle and include the IP address.
 - Connections: Draw arrows or lines to show the direction in which messages travel.

Sample

Router 2	Success	1.1	2.12	What's your favorite color?
Router 6	Success	1.1	2.12	What's your favorite color?
Router 8	Success	1.1	2.12	What's your favorite color?
Router 1	Success	1.1	2.12	What's your favorite color?

Find the message in your router log and record the routers it went to along its path to the destination.



Network Diagram

Part 3: Analyzing Redundancy

1. Identify Redundant Paths:
 - Look at the network diagram you just made. Circle or highlight any instances where messages took different routes to reach the same destination.
2. Scenario Challenge: Router Failure
 - Scenario: Imagine one router fails.
 - Task: On your diagram, draw an alternate route that your message might take to reach the same destination.
 - Explain: In a few sentences, describe how the redundant path prevents a complete network failure:

Helpful Sentence Starters

- I noticed my message traveled through routers _____, _____, and _____.
- If Router _____ failed, the message would _____.
- Redundancy helps by _____.

Challenge:

1. On the back of this paper, sketch a network diagram that has a single point of failure (only one way to reach some devices)
2. Redraw the network paths by adding redundancy so that if any one router fails, the devices still stay connected.

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Lesson 6: HTTP and DNS

Resources

Instructions:

Walk around the room to collect your classmates' IP addresses and create a list.

Rules:

- Share information with only one classmate at a time.
- Make sure to keep all information as up to date as possible.

Name	IP Address

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Lesson 7: The Internet's Hidden Barriers

Resources

Activity Guide - Internet Access IRL

Instructions:

- With your partner, choose one of the personas to interact with
- Ask your persona questions about how they access and use the internet
- Find one graph from the Pew Research data that connects to your persona's experience. Take a screenshot and share it in the chat.
- Record your answers below

Activity:

Which persona did you pick and what did you learn about them?

How does their experience connect to the disparities shown in the Pew Research data?

What is one challenge they face in accessing technology?

Brainstorm one potential solution.

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Lesson 8: Internet Access Advocacy Campaign Project

Resources

Internet Access Advocacy Project - Gallery Walk Info Tracker

Do This:

1. Visit each station to explore real-world challenges related to Internet access.
2. Read the provided scenario or resource at each station and record information below.

Station 1: Diego – High School Student Without Home Internet

1. What challenges does Diego face as a student without home internet access?

2. How might this impact his ability to succeed in school and plan for the future?

3. If you were Diego's school principal, what policies or programs could you put in place to support students like him?

Station 2: Lisa – Small Business Owner in a Rural Area

1. How does unreliable internet access affect Lisa's ability to run her small business?

2. Why might net neutrality be important for small business owners like Lisa?

3. If you were a policymaker, what steps could you take to ensure better internet access for rural businesses?

Internet Access Advocacy Project - Gallery Walk Info Tracker

Station 3: Mr. Johnson – Retired Senior Without Digital Literacy

1. Why do you think so many older adults struggle with digital literacy?

2. How does limited internet access affect Mr. Johnson's ability to manage his healthcare and stay connected with family?

3. What are some barriers that prevent seniors from learning digital skills?

Station 4: Aisha – College Student Affected by Internet Speed Restrictions

1. How does slow or restricted internet access affect Aisha's ability to complete her college assignments?

2. What would happen if universities had to start paying ISPs more money to ensure fast access to learning platforms?

3. If you were Aisha, how would you advocate for stronger net neutrality protections?

Final Reflection: Answer the following questions after visiting all stations:

Which persona's story impacted you the most? Why?

Name(s) _____ Period _____ Date _____

Internet Access Advocacy Project - Gallery Walk Info Tracker

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What are some similarities between the challenges these four individuals face?

What solutions do you think could help close the digital divide and make internet access more equitable for everyone?

Your Name: _____ Presenters' Names:_____

What was the most compelling part of this campaign? <i>(What stood out and made the argument strong?)</i>	
What is one question you have for the presenters? (Something that could make their campaign clearer or stronger?)	
What is one suggestion for improvement? (How could the campaign be more persuasive, actionable, or better supported with evidence?)	

Your Name: _____ Presenters' Names:_____

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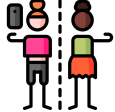
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Social Engineering

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Lesson 10: Emerging Technologies

AI vs Traditional Security - Free Sketch (Optional)

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Question of the Day

What makes AI effective at enhancing network security, and how does it compare to traditional methods?

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Lesson 10: Emerging Technologies

Resources

Activity Guide - Threat Detection Notetaker

Instructions: Choose a card from the Threat Detection Card stack. Read the scenario and ask the Code.org AI Chatbot how AI detected the threat (e.g. *How did AI detect the Phishing threat in the email alert system?*). Ask it for the difference between a traditional response compared to an AI-powered response. Complete the prompts below.

Reference Sheet

Word	Definition
Threat	Something that could damage or break into a system (e.g., malware, scam, breach)
Vulnerability	A weakness that makes a system easier to attack (e.g., outdated software)
Anomaly	Something unusual in network activity that may be a sign of a threat
Phishing	A fake message (email, text) trying to trick someone into giving personal info
Breach	When unauthorized people get access to secure data
False Positive	When AI flags something as a threat—but it's actually safe
AI Bias	When an AI system makes unfair decisions based on incomplete or skewed data
Mitigation	An action or tool that reduces the risk of a threat

Round 1

Card # and Name	
How did AI detect this threat?	

What mitigation strategy does AI recommend?	
How would this threat have been handled without AI?	

Round 2

Card # and Name	
How did AI detect this threat?	
What mitigation strategy does AI recommend?	
How would this threat have been handled without AI?	

Round 3

Card # and Name	
How did AI detect this threat?	
What mitigation strategy does AI recommend?	
How would this threat have been handled without AI?	

Reflection

Instructions: Choose a scenario you completed above. Answer the following questions.

Scenario Name: _____

1. How might the AI response be more effective than traditional methods or responses for this scenario?

2. What challenges might arise when using AI for this kind of network security?